

Gerrymandering Metrics

2024 Election Results and the 2024 District Map

Metrics

Mean-Median approach:

Remember we need to disregard districts in which there was a choice between two democrats or two republicans !

```
district_level <- sr_precinct |>
  group_by(DISTRICT) |>
  summarize(DEM01 = sum(DEM01, na.rm = T),
    DEM02 = sum(DEM02, na.rm = T),
    REP02 = sum(REP02, na.rm = TRUE),
    REP01 = sum(REP01, na.rm = T),
    TOTVOTE = sum(TOTVOTE, na.rm = T)) |>
  mutate(WINNER = ifelse(DEM01 > REP01, "DEM", "REP"),
    SCALE = (DEM01+DEM02)/TOTVOTE)

# Now we can just compute our score :

district_vi <- district_level |>
  filter(DEM02 == 0 & REP02 == 0) |>
  mutate(v_i = DEM01 / (TOTVOTE)) |>
  drop_na(v_i)

mean_median <- district_vi |>
  summarize(democratscore = mean(v_i) - median(v_i))

mean_median
```

```
# A tibble: 1 × 1
  democratscore
      <dbl>
1      -0.00549
```

The mean-median score is of -0.00549 . But we don't really know what to compare it to! It seems there is a very small negative score in defavor of democrats.

Side quest : *Is the mean median score statistically significant ?*

We do a simple bootstrap approach. A deeper analysis would imply a t.test statistic.

```
# We define a function that can compute that systematically on a bootstrap
test:
```

```
democratscore_fn <- function(data, index) {
  d <- data[index, ]
  mean(d$v_i) - median(d$v_i)
}
set.seed(123)
boot_res <- boot(district_vi, democratscore_fn, R = 2000, )

boot.ci(boot_res, type = "perc")
```

BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
Based on 2000 bootstrap replicates

CALL :
boot.ci(boot.out = boot_res, type = "perc")

Intervals :
Level Percentile
95% (-0.0432, 0.0183)
Calculations and Intervals on Original Scale

The CI interval reveals there is no statistical significance on this measure (it contains 0)!

Efficiency Score

```
n_wasted <- function(by, against) {
  w_threshold <- floor((by + against) / 2) + 1
  n_wasted <- ifelse(by > against, by - w_threshold, by)
  n_wasted
}
ES <- district_level |>
mutate(wasted_D = n_wasted(DEM01, REP01),
       wasted_R = n_wasted(REP01, DEM01))|>
summarize(EG = (sum(wasted_R) - sum(wasted_D)) / (sum(TOTVOTE)))

ES
```

```
# A tibble: 1 × 1
  EG
<dbl>
1 0.0881
```

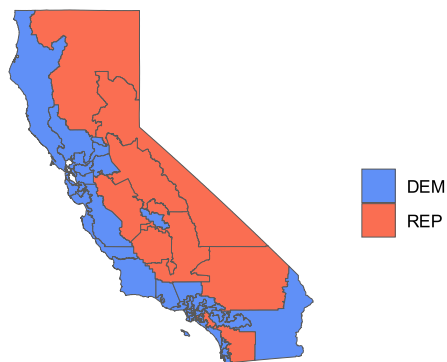
The efficiency score is of 0.08 : A slight advantage in favor of the democrats. ## Map !

```

Reading layer `US_Congressional_Districts' from data source
  `/Users/sachabondeville/Documents/stat133/Projects/Project-5/gerrymandering-
sachabdv1/data/Districts/US_Congressional_Districts.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 52 features and 4 fields
Geometry type: MULTIPOLYGON
Dimension:      XY
Bounding box:   xmin: -13849210 ymin: 3833656 xmax: -12705030 ymax: 5162404
Projected CRS:  WGS 84 / Pseudo-Mercator

```

2024 Gen. Election actual results



Source: Official State data

2024 Election Results and the proposed 2025 District Map

Methodology :

Using data from registered voters in each blocks, we can forecast the new district democrat / republican distribution joining different public datasets. This estimation will then allow us to study the impact of PROP50 redistribution !

```

Reading layer `AB604' from data source
  `/Users/sachabondeville/Documents/stat133/Projects/Project-5/gerrymandering-
sachabdv1/data/AB604/AB604.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 52 features and 15 fields
Geometry type: MULTIPOLYGON
Dimension:      XY
Bounding box:   xmin: -13857270 ymin: 3832931 xmax: -12705030 ymax: 5162404
Projected CRS:  WGS 84 / Pseudo-Mercator

```

```

block_prop <- sr_precinct |>
  mutate(PROP_DEM_BLOCK = (DEM/sum(DEM))*(PCTSRPREC/100),

```

```
PROP_REP_BLOCK = (REP/(sum(REP)))*(PCTSRPREC/100),
.by = SRPREC_KEY)|>
mutate(BLOCK_WEIGHTED_DEM = (DEM01+DEM02)*PROP_DEM_BLOCK,
BLOCK_WEIGHTED_REP = (REP01+REP02)*PROP_REP_BLOCK)
```

Now we have to redistribute these blocks with the new district mapping.

```
hypo_districts <- block_prop |>
  group_by(NEWDISTRICT)|>
  summarize(DEM_est = sum(BLOCK_WEIGHTED_DEM, na.rm = T),
  REP_est = sum(BLOCK_WEIGHTED_REP, na.rm = T))|>
  mutate(WINNER= ifelse(DEM_est > REP_est, "DEM", "REP"))|>
  drop_na(NEWDISTRICT)
```

Metrics :

Mean-Median approach:

Following the same methodology as above :

```
hypo_district_vi <- hypo_districts |>
  mutate(v_i = DEM_est / (REP_est + DEM_est))

hypo_mean_median <- hypo_district_vi |>
  summarize(democratscore = mean(v_i)-median(v_i))

hypo_mean_median
```

```
# A tibble: 1 × 1
  democratscore
      <dbl>
1         0.0254
```

The mean-median score increased slightly with Prop50, this time in favor of democrats.

Side quest : *Is the mean median score statistically significant ?*

```
set.seed(43)
boot_res_2 <- boot(hypo_district_vi, democratscore_fn, R = 2000)

boot.ci(boot_res_2, type = "perc")
```

BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
Based on 2000 bootstrap replicates

CALL :

```
boot.ci(boot.out = boot_res_2, type = "perc")
```

```
Intervals :
Level      Percentile
95%      (-0.0132,  0.0571 )
Calculations and Intervals on Original Scale
```

The mean-median score increased in favor of the democrats, but the result is still not exactly statistically significant. This reveals the weakness of that measure, as PROP50 *intentionally* redistricted, which should have affected the metric more.

Efficiency Score :

```
hypo_ES <- hypo_districts |>
mutate(wasted_D = n_wasted(DEM_est, REP_est),
       wasted_R = n_wasted(REP_est, DEM_est)) |>
summarize(EG = (sum(wasted_R) - sum(wasted_D)) / (sum(DEM_est + REP_est)))

hypo_ES
```

```
# A tibble: 1 × 1
  EG
<dbl>
1 0.214
```

The efficiency gap significantly increased, from 8 to 20%. This metric seems to reveal the impact of prop50 in a more consistent way!

Hypothetical map :

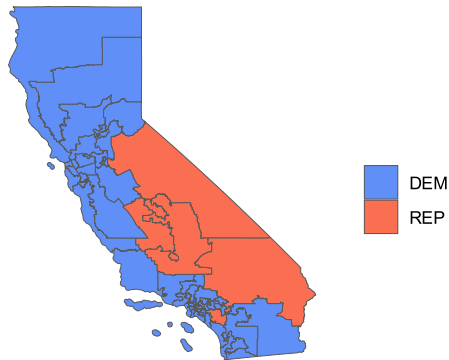
```
hypo_districts <- st_sf(left_join(hypo_districts, ab604, by =
join_by(NEWDISTRICT == DISTRICT)))

hypo_districts <- st_transform(hypo_districts, crs = 4326)

hypo_districts |>
ggplot(aes(fill = WINNER))+geom_sf()+
scale_fill_manual(
values = c(
  "DEM" = "#6090F7",
  "REP" = "#FA6E52"))+
theme_minimal(base_family = "Arial")+
scale_x_continuous(breaks = c())+
scale_y_continuous(breaks = c())+
labs(title = "2024 Gen. Election hypothetical results",
x = "",
```

```
y = "",
fill = "",
caption = "Source: Official State data")
```

2024 Gen. Election hypothetical results



Source: Official State data

```
hypo_districts <- st_transform(hypo_districts, crs = 4326)

#pal_continuous <- colorNumeric(
# palette = colorRampPalette(c("#ff0000ff", "#000dffff"))(100),
# domain = districts$SCALE
#)

pal_discrete_2 <- colorFactor(
palette = c(
  "DEM" = "#6090F7",
  "REP" = "#FA6E52"
),
domain = hypo_districts$WINNER
)
```

Interactive maps code :

```
districts <- st_transform(districts, crs = 4326)

#pal_continuous <- colorNumeric(
# palette = colorRampPalette(c("#ff0000ff", "#000dffff"))(100),
# domain = districts$SCALE
#)

pal_discrete <- colorFactor(
palette = c(
```

```

    "DEM" = "#6090F7",
    "REP" = "#FA6E52"
  ),
  domain = districts$WINNER
)

# Actual results:
leaflet(districts) |>
  addProviderTiles("CartoDB.Positron")|>
  addPolygons(
    fillColor = ~pal_discrete(WINNER),
    color = "black",
    weight = 1,
    fillOpacity = 0.8,
    label = ~paste0("DEM Votes:", DEM01, " | REP votes : ", REP01))|>
  addLegend(
    pal = pal_discrete,
    values = ~WINNER,
    title = "Election result",
    position = "bottomright"
  )

#Hypothetical results:
leaflet(hypo_districts) |>
  addProviderTiles("CartoDB.Positron")|>
  addPolygons(
    fillColor = ~pal_discrete(WINNER),
    color = "black",
    weight = 1,
    fillOpacity = 0.8,
    label = ~paste0("DEM Votes:", floor(DEM_est), " | REP votes : ",
      floor(REP_est))|>
  addLegend(
    pal = pal_discrete_2,
    values = ~WINNER,
    title = "Election result",
    position = "bottomright"
  )

```